COMPUTATIONAL FLUID DYNAMICS

VII Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6AE39	PCC	3	-	-	3	40	60	100

UNIT-I INTRODUCTION TO COMPUTATIONAL FLUID DYNAMICS

Introduction to computational fluid dynamics, Research tool, Design Tool, Finite control volume, infinitesimal fluid element, substantial derivatives, divergence of Velocity. Governing Equations.

UNIT-II GOVERNING EQUATIONS OF FLUID DYNAMICS

Form of Governing equation suited for CFD. Conservation form, Non Conservation forms, shock fitting and shock capturing.Classification of Quasi-Linear Partial differential equation, The Eigen value method, General behavior of different classes of Partial differential equation, elliptic, parabolic and hyperbolic.

UNIT-III DISCRETIZATION AND TRANSFORMATIONS

Introduction, Finite differences and formulas for first and second derivatives, difference equations, Explicit and implicit approaches, multidimensional finite difference formulas, finite difference formulas on non-uniform grids. Problems on 1D , 2D discretization using FDE,BDE,CDE.

UNIT-IV GRID GENERATION

Types of grids, generation. Structured grids- Cartesian grids, stretched (compressed) grids, body fitted structured grids, Multi-block grids - overset grids with applications. Unstructured grids- triangular/ tetrahedral cells, hybrid grids, quadrilateral/hexahedra cells. Grid Generation techniques - Delaunay triangulation, Advance font method. Surface and volume estimations, grid quality and best practice guidelines.

UNIT-V CFD TECHNIQUES

Lax-Wendroff technique, MacCormack's technique, Crank Nicholson technique, Relaxation technique- aspects of numerical dissipation and dispersion, Alternating-Direction-Implicit (ADI) Technique. Pressure correction technique Numerical procedures- SIMPLE, SIMPLER algorithms SIMPLEC and PISO algorithms Boundary conditions for the pressure correction method. Parallel Computing.

Text Books:

1.John .D. Anderson "Computational Fluid Dynamics", McGraw Hill

2.Hoffmann, K.A: Computational Fluid Dynamics for Engineers, Engineering Education System, Austin, Tex., 1989

Reference Books:

1. J Blazek, "Computational Fluid Dynamics: Principles and Applications" Elsevier.
2. Chow CY," Introduction to Computational Fluid Dynamics", John Wiley, 1979

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. Apply the concepts of finite control volume and infinitesimal fluid elements, comprehend substantial derivatives, and interpret divergence of velocity in the context of fluid dynamics.

2. Differentiate between conservation and non-conservation forms of governing equations suitable for CFD, focusing on shock fitting and shock capturing methods.
3. Implement finite difference methods for discretization in 1D and 2D problems, demonstrating proficiency in Finite Difference Equations (FDE),
4. Apply grid generation techniques such as Delaunay triangulation and Advanced Front Method, estimating surface and volume parameters
5. Apply pressure correction techniques and numerical procedures such as SIMPLE, SIMPLER, SIMPLEC, and PISO algorithms, including boundary conditions for pressure correction methods in CFD simulations.